

The shapes of meaning: Diagrams in lexical semantics¹

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Abstract Diagrams in lexical semantics come in all shapes and sizes, but there is currently no good understanding of the design space that allows for this variety. This paper derives that space from the combination of a small number of graphic and semantic parameters, thereby contributing to our insight in this increasingly common visual metalanguage in linguistics and related disciplines concerned with meaning.

Visualization, showing information through diagrams, plays an important role in many scientific disciplines. Linguistics is no exception, but it is especially in studies of word meaning that diagrams are prominent and increasingly so (Section 1). However, despite this prominence and despite a growing literature on linguistic diagrams in general, the reflection about how semantics can be made visible is still restricted and unsystematic, leaving it unclear why word meanings can be visualized in such a variety of ways (Section 2). The contribution of this paper is a descriptive two-dimensional typology of word meaning diagrams, combining four types from the literature on diagrams and visualization (Section 3) with two parameters familiar from semantics (Section 4), creating room for further explorations (Section 5).

1. The visual turn in studies of word meaning

A word meaning diagram can be defined as:

a non-textual and non-pictorial graphic representation of how the meaning or meanings of one or more individual words is/are structured or exemplified.

Let us consider the parts of this definition one by one. A *graphic representation* (or simply a *graphic*) is ‘a visible artifact on a more or less flat surface, that was created in order to express

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information' (Engelhardt, 2002, p. 2). Texts and pictures count as graphics too, hence the adjectives *non-textual* and *non-pictorial* to single out the kind of graphics that we usually call diagrams. (This negative characterization gives way to a positive one in Section 3 and is broadened to include pictures and texts in Section 5.) *Individual words* are not just words in the usual sense, but all elementary meaningful building blocks in language, including those smaller than words (morphemes, affixes, grammatical categories, like case or tense) or larger (multi-word expressions, idioms). The term *meaning(s)* (singular or plural) can be replaced by *sense, function, use, content, referent, definition, denotation*, whatever counts as the semantic side of a word in a particular diagram. Finally, how the meanings of one or more words are *structured or exemplified* is a provisional characterization of the functions of word meaning diagrams that will have to suffice for now but will be amply illustrated in the remainder of this section.

Word meaning diagrams have been used for ages (Section 1.1), but there was what we might call a 'visual turn' in studies of word meaning in the decades around the beginning of the 21st century (Section 1.2). Before presenting a small but representative sample of the many word meaning diagrams found in the literature from these two stages, three remarks are in order.

First, the examples from the literature are adapted in a systematic way, for expository reasons. A word meaning diagram is always (explicitly or implicitly) about one or more words. When they occur in the diagram, adaptation has those words in a black, bold, italicized serif font. Everything else is in grey, with any text that does not mention object words in a regular roman sans serif font. This will help us to see more clearly where the words are and where the meanings are in word meaning diagrams. The adaptations do not affect the contents of the originals, but colors, artistic style, and minor details had to be sacrificed sometimes.

Second, this paper is about how word meaning diagrams work, not about the (empirical or theoretical) correctness of what they present, or the attractiveness or effectiveness of how they do it. This is a descriptive study, without prescriptive intentions. Also, all word meaning diagrams are potentially relevant, whether old or young, small or big, hand-made or computer-assisted, sophisticated or clumsy.

Third, with so many different diagrams it is impossible to explain the contents of each diagram in detail. Many diagrams speak for themselves or show enough of their relevance for this paper without the need to get into their substance. The caption always has a specific reference to the source, for readers who are curious for more background.

1.1 Older word meaning diagrams²

Aristotle's square of oppositions (Figure 1) and Porphyry's tree (Verboon, 2014) can be seen as the earliest word meaning diagrams, originating, despite their Greek eponyms, in medieval philosophy. In the 18th and 19th century, mathematicians and philosophers started to use circles for categories and their relations, for quantificational determiners (Figure 2) and cross-linguistic lexical differences (e.g., in Schopenhauer's work, see Wälchli (2010, p. 336) and Lemanski (2020)). In the 20th century, starting with Saussure (1916), structural linguistics brought along a rich variety of word meaning diagrams, for instance, in the work of Hjelmslev (1943, pp. 49, 50), Guiraud (1955, p. 72; based on Trier, 1934), Jakobson (1958, p. 126), Coseriu (Figure 3), Pottier (1964), Lehrer (Figure 4).³

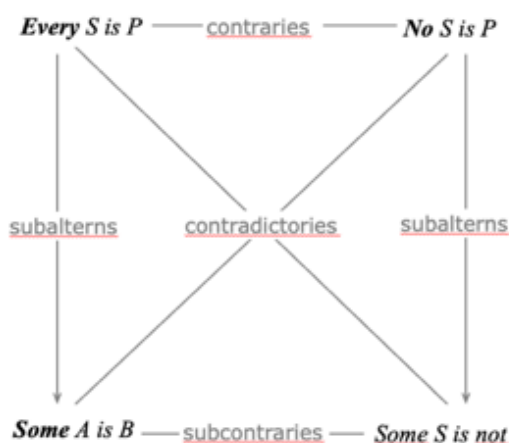


Figure 1: Square of oppositions between *every*, *some*, *no* (Parsons & Ciola, 2025)

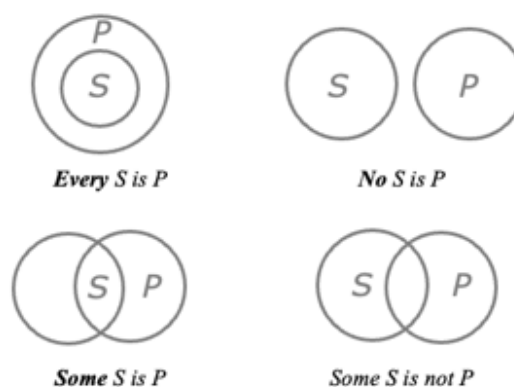


Figure 2: Euler's diagram for *No S is P* (Lemanski, 2020, p. 50)

	'old'	'not old'
'for persons'	<i>senex</i>	<i>iuvenis</i>
'for animals and plants'	<i>vetulus</i>	<i>novellus</i>
'for things'	<i>vetus</i>	<i>novus</i>

Figure 3: Latin 'age' words (Coseriu, 1964, p. 151)

<i>cook</i>								
<i>boil</i>		<i>fry</i>		<i>broil</i>		<i>bake</i>		
<i>simmer</i>	<i>boil</i>	<i>sauté</i>	<i>deep-dry</i>	<i>grill</i>	<i>barbecue</i>	<i>plank</i>	<i>roast</i>	<i>scallop</i>

Figure 4: Cooking verbs (Lehrer, 1969, p. 45)

² See D'Andrade (1995) and Geeraerts (2010) for relevant historical overviews, with many word meaning diagrams, from anthropology and linguistic semantics, respectively.

³ Some of these diagrams have generated their own secondary literature: Christensen (2016) and Willems (2018), for instance, about Saussure's diagrams, Cigana & Polis (2023) about Hjelmslev's, Chvany (1984, 1986) and Lauersdorf (2018) about Jakobson's.

Partially in parallel, sometimes with larger data sets and more sophisticated methods, anthropologists and psychologists used various visualizations in their studies of connotations (Osgood, 1952, p. 229), kinship (Romney & D'Andrade, 1964), folk taxonomies (Berlin et al., 1968), address systems (Ervin-Tripp, 1969, p. 101), body parts (Friedrich & Krauss, 1969, p. 38), emotions (Russell, 1980, p. 1167), colors (Figure 5), to name a few examples.

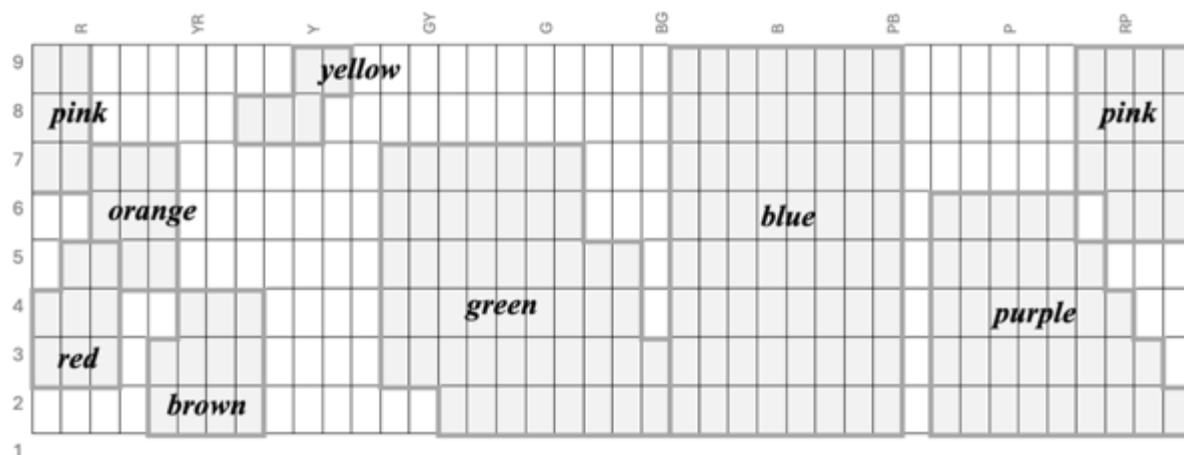


Figure 5: American English basic color terms (chromatic) on the Munsell color chart (Berlin & Kay, 1969, p. 119)

Other well-known word meaning diagrams from this earlier period are Reichenbach's tense diagrams (Figure 6), the decompositional trees of *bachelor* in Katz and Fodor (1963, p. 186), the semantic bird network in Collins and Quillian (1969, p. 241), and the visuals in Labov's (1973) work on the boundaries of *cup*.



Figure 6: Order of R(eference), E(vent), S(peech) time in *Simple Past* (Reichenbach, 1947, p. 290)

1.2 Newer word meaning diagrams

In the decades around the millennium a number of different (sometimes overlapping) developments contributed to what we might call a visual turn in the study of word meaning, with a more central role for diagrams.

Cognitive linguistics The emergence of cognitive linguistics in the 1980s (Fauconnier, 1985; Talmy, 1985; Langacker, 1986; Lakoff, 1987; Johnson, 1987; Geeraerts, 1989) was accompanied by a prominent role for various types of diagrams, like the radial network for prototype categories, e.g., Figure 7.

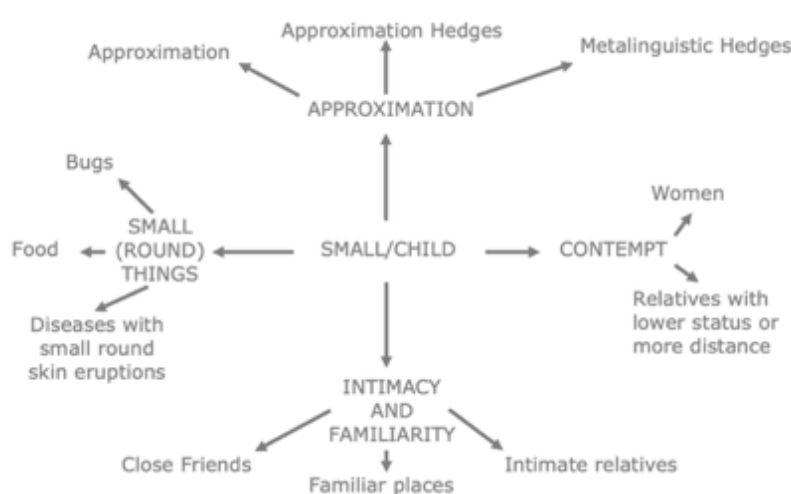


Figure 7: Meanings of Cantonese raised tone diminutive (Jurafsky, 1996, p. 539)

Semantic maps Semantic maps find their origin in cross-linguistic studies of tense, voice, indefiniteness, modality, as hand-made networks of meanings in which polysemous words or grammatical markers form contiguous regions (L. B. Anderson, 1982; Kemmer, 1993; Haspelmath, 1997; Van der Auwera & Plungian, 1998; Croft, 2001). Figure 8 is an illustration of this type.⁴ Later versions used computational methods (e.g. Regier et al., 2013), often with scaling techniques (Croft & Poole, 2008; van der Klis & Tellings, 2022), with data that is

⁴ The network of functions of such a classical semantic map is constructed, with as few links as possible, in such a way that the subset of functions of every multifunctional item in the cross-linguistic data set forms a contiguous (i.e., connected) sub-network.

experimentally elicited (e.g., Majid et al., 2007) or taken from parallel corpora, e.g., van der Klis et al. (2022) and Figure 9.^{5,6}

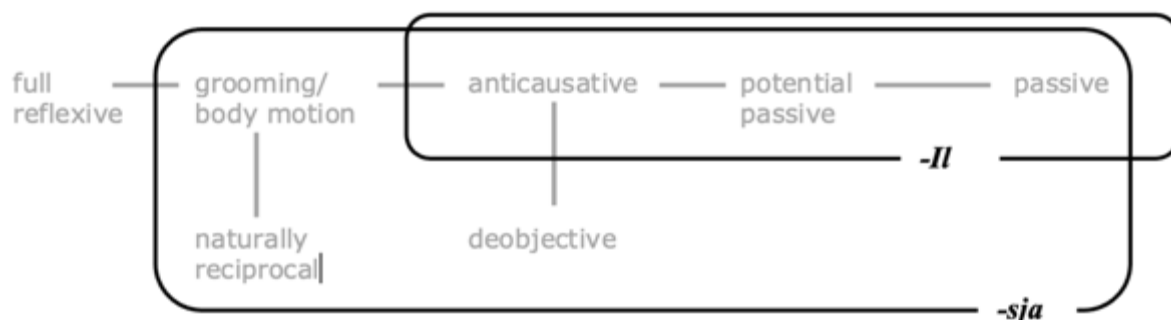


Figure 8: Distribution of Russian *-sja* and Turkish *-II* over reflexive/middle functions (Haspelmath, 2003, pp. 225–226)

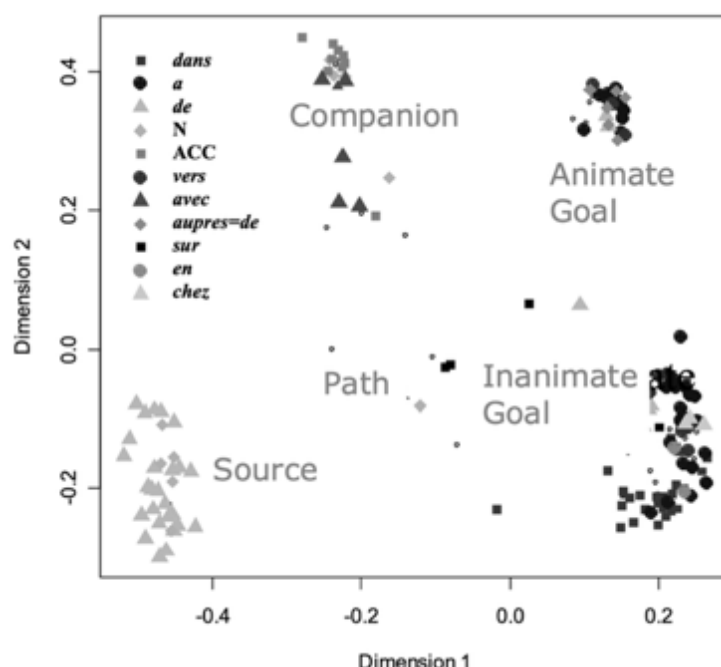


Figure 9: Prepositions in motion clauses in a French translation of the Gospel of Mark (Wälchli, 2010, p. 348)

Naming studies The decades around 2000 show an intensification of systematic (typically comparative) studies eliciting names of visual stimuli and analyzing the data with statistical scaling techniques. Some examples of domains that this ‘extensional approach’ is applied to,

⁵ The 190 data points correspond to motion clauses in the gospel of Mark. On the basis of translations in 153 different languages, multidimensional scaling is used to define a similarity space in which two clauses are closer together if languages use the same prepositions more often in those two clauses.

⁶ See, among others, Haspelmath (2003), Van der Auwera (2013), Georgakopoulos and Polis (2018, 2022) for general overviews and discussions.

with a variety of diagrams, are containers (Malt et al., 1999), topological relations (Levinson & Meira, 2003), body parts (Enfield et al., 2006), causation (Wolff, 2007), cutting and breaking events (Majid et al., 2007), human locomotion (Malt et al., 2008), and reciprocity (Evans et al., 2011). The linear diagram Figure 10 is a relatively simple illustration from this literature.

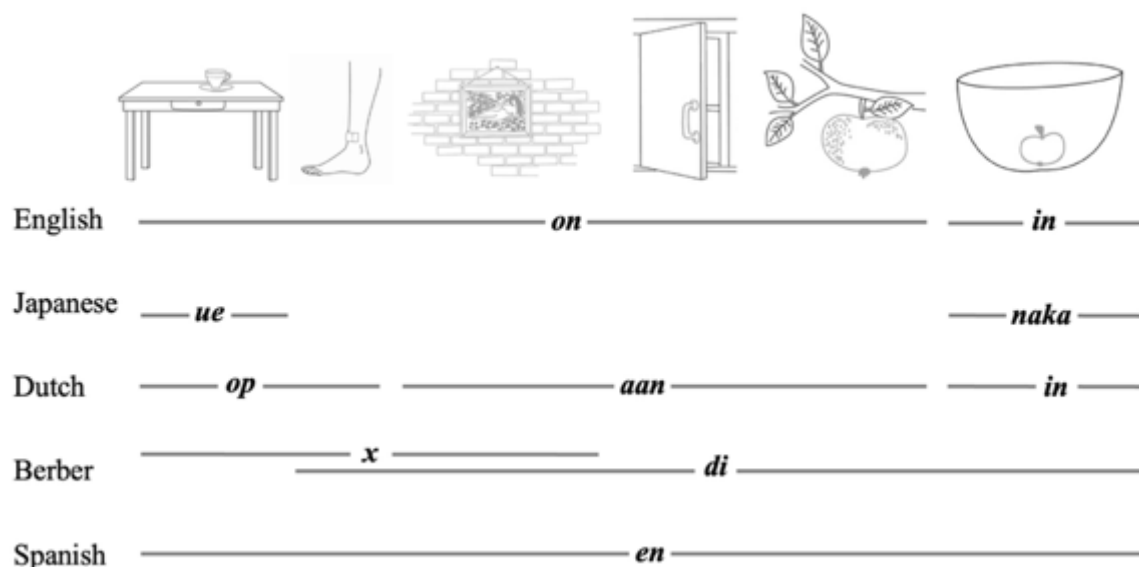


Figure 10: Differences in categorizing spatial relations (Bowerman & Choi, 2001, p. 485)

Lexical databases The availability and accessibility of large lexical databases in electronic format is another factor in the visual turn. We find colorful, often interactive visualizations, for instance, of Roget's thesaurus (Old, 2002), of colexified concepts in the CLICS database (Mayer et al., 2014),⁷ metaphors in the Oxford English Dictionary (W. Anderson et al., 2016), and WordNet (Figure 11).

⁷ CLICS stands for 'cross-linguistic colexifications'. Two senses or concepts are colexified if the same lexical form is used for them (A. François, 2008, p. 170), the phenomenon that also underlies semantic maps.

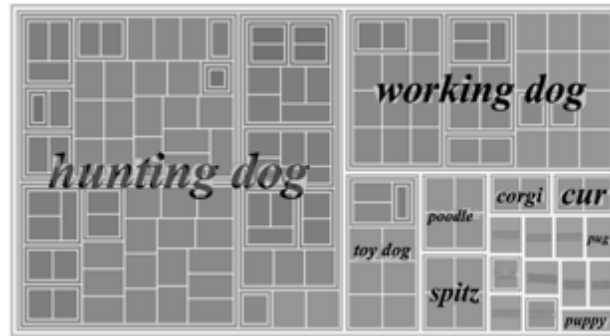


Table 11: Treemap of the WordNet hyponyms of the domestic animal sense of *dog* (C. Collins & Penn, 2007, p. 4)⁸

Frame semantics Visual networks play an important role in approaches that see word meanings and concepts as organized in terms of frames, i.e., larger structures of conceptual and encyclopedic information (Barsalou, 1992; Fillmore & Baker, 2015; Löbner, 2021).

Figure 12 is an example that shows the parts and properties characteristic of *pencil*.

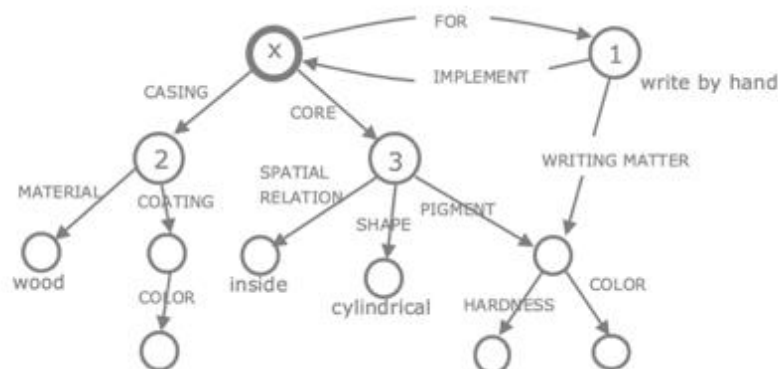


Figure 12: Frame for *pencil* (Löbner, 2021, p. 265)

Features A different approach to the building blocks (features) of word meanings uses attribute-value pairs or matrices (AVMs), e.g., Copestake (1992) and Pustejovsky (1995), with the typical recursive bracketings found in Figure 13, for instance.

⁸ The treemap is a hierarchical structure in which rectangles are divided into smaller rectangles. Figure 11 shows the immediate hyponyms of *dog*, with larger rectangles for words that have more hyponyms themselves. Only some of the words corresponding to the boxes are shown.

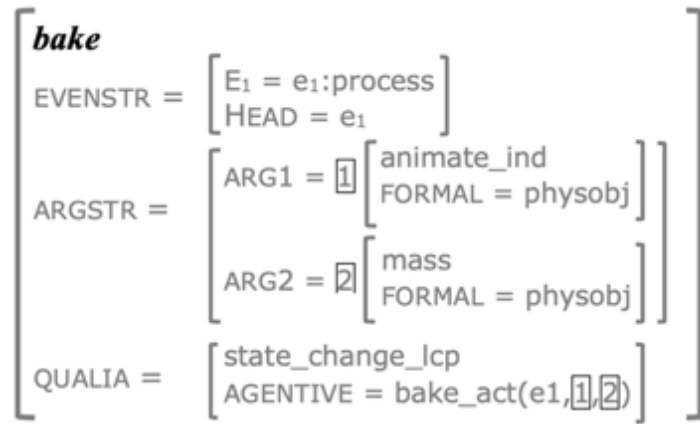


Figure 13: The lexical structure of *bake* (Pustejovsky, 1995, p. 123)

Distributional semantics The distributional hypothesis (Sahlgren, 2008) says that words with similar meanings occur in similar contexts, i.e., have similar distributions in large corpora. These distributions can be modeled as word vectors in a high-dimensional space and their similarities visualized through spatial proximities (Figure 14). This method also allows the modeling of proximities within a set of tokens (individual occurrences) of one word in a corpus, allowing potential insight in the different uses of a word (Figure 15).

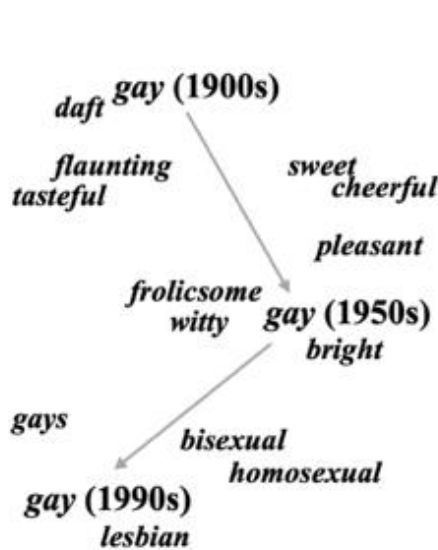


Figure 14: Semantic shift of *gay* (Hamilton et al., 2016, p. 1490)

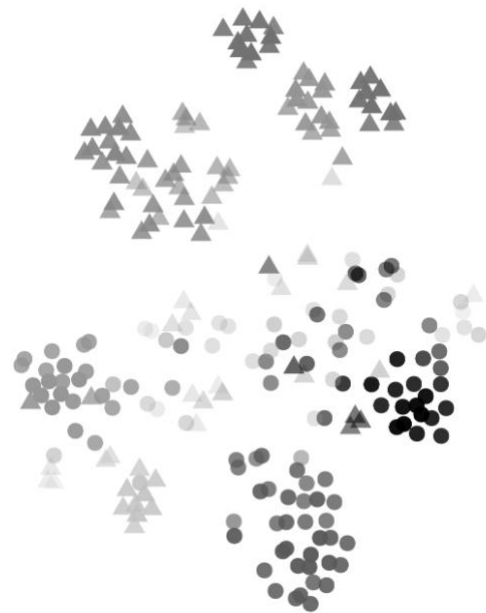


Figure 15: Uses of Dutch *hachelijk* ‘dangerous, critical’ with different nouns (adapted from Montes, 2021, p. 119)

Explaining semantics There has been no lack of semantics textbooks appearing over the past decades. Some of these make abundant use of diagrams to explain semantic notions or insights in an intuitive way (e.g., Geeraerts, 2010; Murphy, 2010; Goddard, 2011; Löbner, 2013; Riemer, 2016; Valenzuela, 2017; Feist, 2022; Lee, 2023). In this context we find simple word meaning diagrams with a deliberate expository purpose, like the two ways to illustrate prototypicality in the domain of fruit in Figure 16 and 17.⁹

	of wood-plant	juicy	sweet
Apple	+	+	+
Strawberry	-	+	+
Banana	+	-	+
Orange	+	+	-

Figure 16: Prototypicality of *fruit* (adapted from Geeraerts, 2010, p. 191)

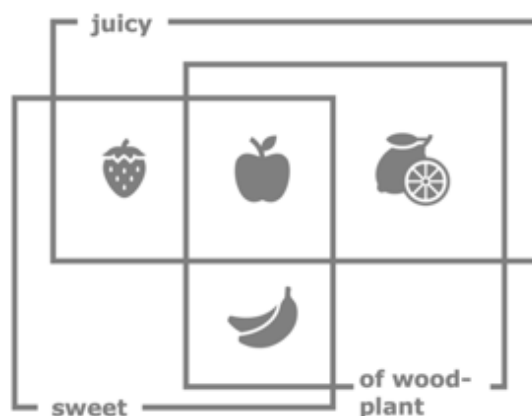


Figure 17: Prototypicality of *fruit* (adapted from Geeraerts, 2010, p. 191)

Geometry, space and meaning Beyond the visual, expository level there is in the visual turn often a deeper concern with the underlying geometrical organization of meaning in language. The word *geometry* appears in quite a few titles related to meaning in the last few decades and in all these publications diagrams play an important role:

‘Geometric representation of paradigms in a modular theory of grammar’ (McCreight & Chvany, 1991), *Conceptual spaces: The geometry of thought* (Gärdenfors, 2000), ‘The geometry of grammatical meaning’ (Haspelmath, 2003), *Geometry and meaning* (Widdows, 2004), *The geometry of meaning: Semantics based on conceptual spaces* (Gärdenfors, 2014), ‘Logical geometries and information in the square of opposition’ (Smessaert & Demey, 2014), *Language, space and mind: The conceptual geometry of*

⁹ The two diagrams are made even simpler than the originals by leaving only three of the five original attributes.

linguistic meaning (Chilton, 2014), ‘The geometry of linguistic oppositions and the issue of paradigmatic contrast’ (Cigana, 2024)

The term *geometry* might be metaphorical for some, but for Gärdenfors, in particular, it appropriately fits in a broader vision of linguistic meaning as rooted in the spatial organization of human cognition, with conceptual dimensions and distances and with *convexity* as a constraint on the shape of concepts and word meanings. While geometry in Gärdenfors’ vision is rooted in cognition, scholars of logical geometries, like Demey and Smessaert, are after the deep formal foundations shared by geometry and logic. The diagrams that Gärdenfors and Demey and Smessaert present are not just expository tools, but their spatial and geometric nature is close to the cognitive and logical realities that they are intended to represent.

Having seen this variety of word meaning diagrams we now move to the meta-level of studies *about* linguistic diagrams. How can these help us to understand this variety?

2. Visual linguistics

The term *visual linguistics* (Bubenhof, 2018, 2020) can be used to refer to all efforts to understand and develop diagrammatic representations of linguistic information, that is, information about language and linguistics. This is not a prominent department within linguistics, but it is growing, with dissertations (C. Collins, 2010; Montes, 2021; Rohrdantz, 2014; Schätzle, 2018), edited volumes (Bubenhof & Kupietz, 2018; Butt et al., 2020; Mazziotta et al., 2023b), an encyclopedia article (Mazziotta, 2026), and an expanding list of papers that is already getting too big to review here.

Since the goal of this paper is a typology of word meaning diagrams, linguistics literature is singled out that might have something to offer for such a typology: general classifications of linguistic diagrams (Stewart, 1976; Smessaert & Demey, 2018; Petricca, 2019; Bubenhof, 2020) and discussions of formats for lexical-semantic information (Kay, 1971; Vander Meulen, 1990; Berlin, 1992; Geeraerts, 1995, 2007; Croft, 2022).¹⁰

¹⁰ Some of the potentially relevant scholarship on ‘visual linguistics’ is written in French (Badir, 2023; Badir & Polis, 2023; Col, 2023; J. François, 2022, 2023; Mazziotta et al., 2023a, 2023b, 2023c) and was therefore unfortunately not accessible to me.

Smessaert & Demey (2018) make a useful overall distinction between *linguistic parameters* (*what* is represented, the *content*) and *diagrammatic parameters* (*how* it is represented, the *form*). The content of a diagram according to Smessaert & Demey can be:

- monolingual or multilingual
- static (synchronic) or dynamic (diachronic)
- monomodal (e.g., only semantics) or multimodal (e.g., semantics and syntax)
- object-level ('expressions and their properties', p. 239) or metalevel ('properties of, or relations between, the linguistic concepts or the applied terminology', p. 239)
- qualitative or quantitative
- monodisciplinary (only linguistics) or interdisciplinary (also other disciplines)

Word meaning diagrams are, by definition, always monomodal (only semantics) and object-level (about words and their meanings), but they can vary with respect to all the other parameters. Most are monolingual, but some are multilingual (e.g., Figure 8). Most are static, but they can be dynamic (Figure 14). Most are qualitative, some are quantitative (e.g., Figure 9). Most are monodisciplinary, but Figure 5 brings in the science of colors. So, most of these linguistic parameters apply to word meaning diagrams, but that is not enough: if we want to understand the variety among word meaning diagrams in particular, what we need are specific *semantic* parameters (see Section 4).

For the diagrammatic dimension, Smessaert & Demey distinguish *iconic* diagrams (pictures and maps) and *symbolic* diagrams, which are further divided into *charts* ('numerical or quantitative information'), *graphs* ('lines to depict relationships', p. 240), and *set diagrams* ('overlapping geometric shapes'). They also mention *tables* elsewhere in their paper, once together with text and notation as a 'linguistic form' (p. 237) and once together with charts, as a way to represent quantitative information (p. 239), so the status of tables remains unclear.¹¹

Decades before Smessaert & Demey, Stewart (1976) presented a classification of diagrams in phonology, syntax, semantics and historical linguistics, recognizing *tree diagrams*, *matrix diagrams*, and *box diagrams*. Trees are the classical node-link diagrams of

¹¹ The other diagrammatic parameter is *static* versus *dynamic*. Diagrams are usually static ('no graphical elements representing a succession or sequence', p. 241), but can be dynamic ('a succession or sequence of states or processes, either as a series of individual diagrams, or by using graphical elements such as arrows', p. 241). In our sample, only Figure 14 is clearly dynamic, because of the arrows (and could also have been represented graphically as a sequence of three panels with *gay* at different positions in the same constellation of background words).

hierarchical structure. Matrices are essentially tables. The category of box diagrams includes hierarchical structures of stacked or nested rectangles with linguistic elements inside them (stacking in Figure 4, nesting in Figure 11) but also diagrams that have linguistic elements at the vertices of squares and cubes and binary contrasts along the edges or diagonals (like Figure 1). Petricca (2019) distinguishes a *hierarchical structure* (‘records organized in a tree structure’, p. 485), a *relational structure* (‘typically represented as a table’, p. 485), and an *object-oriented model* (which he illustrates with a diagram from Schalley (2004, p. 3) for the verb *wake up* using the Unified Modeling Language). Bubenhofer (2020) has *lists* (including tables), *maps* (in the true geographical sense), *partituras* (multi-tiered ‘scores’ of a stretch of discourse or speech), *vector* diagrams (points in a coordinate system), *graphs* (network).

Kay (1971) and Berlin (1992) discuss how *tree diagrams*, *box diagrams*, and *Venn diagrams* can all be used to represent ethnobiological hierarchies of words, with the words corresponding to circles in a Venn diagram, nodes in a tree diagram, boxes in a box diagram. In a tree diagram one mother node is connected to the daughter nodes below it by lines, in a box diagram a mother box is stacked on top of its daughter boxes, and in a Venn diagram a mother circle includes its daughter circles. Vander Meulen (1990) discusses the usefulness of Venn diagrams for representing meaning relations between words. Geeraerts (1995) is about different visualizations of the prototypical structuring of the exemplars or senses of one word. He distinguishes an *overlapping sets model* (like Figure 17) and a *radial network model* (like Figure 7), and tables are shown for traditional componential or field analyses (like Figure 3). Geeraerts (2007) adds a *multidimensional model*, a plot with multiple axes (like Figure 15). In each case, the whole diagram corresponds to one word, the elements or nodes are instances of the word, and components (features) are visible as sets, links, or axes. Finally, in Croft (2022), and other publications about semantic maps, a distinction is made between ‘connectivity maps’ (*graphs* in mathematical terminology), where the meanings are nodes in a network, and ‘proximity maps’, where the meanings are points plotted in a Euclidean space. In both cases words are second order, as sets of meanings.

This overview shows that the diagram types of *networks*, *plots*, and *set diagrams* figure prominently in this literature, but *tables* less so and with unclear definitions. We do not know what kind of diagram a box diagram is and it is also not clear how trees fit into the typology, given that they can take the shape of networks, set diagrams, and box diagrams. Authors usually give only *lists* of diagram types, not organized by an overarching diagrammatic framework. We also saw that words and meanings can relate to diagrams in different ways. The nodes of a tree can represent words, but also meanings; the circles of a set

diagram can be words, but also semantic features. It is unclear what kind of mappings we see in general between linguistic content and diagrammatic form. What is needed is an integration of these earlier insights into a two-dimensional typology of word meaning diagrams, in which the graphic dimension is more than an open list of intuitive types and there are parameters specifically for the semantic dimension.

3. The graphic dimension

Given the sheer size of research on diagramming and visualization only a few scholars can be mentioned here, because they were particularly helpful for the typology of this section: Horn (1998), Börner et al. (2015; 2019), and Engelhardt and Richards (e.g., 2020, 2024). What emerges from these sources is a small number of types of elementary building blocks ('morphological units'/'primitives' (Horn), 'symbols' (Börner), 'components' (Engelhardt & Richards)) together with a small number of ways of combining these ('syntax'/'topologies' (Horn), 'visualization types' (Börner), 'encodings' (Engelhardt & Richards)).¹²

These are the three basic types of graphic components:

- *pictures* are iconically related to their content, by visuospatial resemblance, involving a form of spatial projection, with more fidelity (more mimetic or realistic) or less fidelity (more schematic or stylized),¹³
- *texts* (consisting of letters, digits, and punctuation) are symbolic representations of linguistic expressions (of sounds, words etc.),
- *shapes* are elementary geometric figures (lines, arrows, circles, dots, triangles, curves, ...) with symbolic functions (for relations, categories, elements, properties).

These components combine in a graphic space (the 'canvas') to form different types of diagrams. Simplifying somewhat, we can think of diagrams as having a first-order level (a 'universe') of elementary graphic individuals (of one of the three types above) on which the diagram imposes a second-level structure of relations, categories, and properties through its geometric shapes and spatial arrangements.

¹² Important for graphics in general, but less for our typology are the so-called *graphic variables* (Bertin, 1983), properties of size and color, for instance, that can be varied for meaningful purposes, as in Figure 9 and Figure 15.

¹³ There is much more to say about pictures, iconicity, and projection, of course. See, for instance, Greenberg (2011).

- A *network* draws *links* between individuals (as *nodes*). These links can be non-directed or directed (arrows) and vary in other ways (e.g., thickness).
- A *plot* positions individuals (as *points*) along one or more *coordinate axes*. These axes can be explicitly represented by lines but also be implicitly understood from the dimensions of the canvas.
- A *set diagram* draws *boundaries* along, around or between the individuals (as *members*). The three prepositions (*along*, *around*, *between*) are needed to cover not only regular set diagrams (boundaries around a set), but also *linear* diagrams (line segment along a set) and line segments separating the members of two sets.
- A *table* positions individuals (as *entries*) in a *grid*, a partition of a space into geometric shapes, the cells, without overlap and without gaps.

In each case there is a specific graphic structure (with axes, cells, boundaries, links) of a set of graphic individuals (functioning as points, entries, members, nodes).

Plots and tables are similar in organizing a whole graphic space and thereby constraining the position of their individuals (tables a bit less than plots, because the position of an entry in a table is only constrained by the cell that it is in). In contrast, networks and set diagrams have a ‘rubber-sheet geometry’: their nodes and members enjoy a relative freedom of position, as long as the ‘topology’ of links and boundaries is maintained. This corresponds to the possibilities of combining graphic types: the geometries of plots and tables exclude each other, but boundaries and links can always be combined with any other graphic type.

Networks The networks are Figures 1, 7, 8, and 12. Networks can be oriented in a particular way (e.g., radial in Figure 7, more or less vertical in Figure 12) or form a geometric shape (e.g., square in Figure 1, cube in Jakobson (1958)). In all these examples the nodes are texts, but they can also be pictures (e.g., in the semantic map of topological relations in Regier et al. (2013)) or shapes with texts (e.g., in Ervin-Tripp’s (1969) flowcharts for terms of address).

Plots The plots are Figures 6, 9, 14, and 15. There is variation in the number of axes, how these are arranged, and how explicit or precise they are. The points can be shapes, texts (or combinations of them), or pictures (e.g. in the MDS plots of topological relations in Levinson & Meira (2003)).

Set diagrams Figures 2, 10, and 17 are set diagrams. The boundaries can have different shapes (circular, rectangular, linear, respectively). They can work without members, to indicate relations between classes (Figure 2). The members of Figure 10 and 17 are pictures but set diagrams with words as members are possible as well, of course (e.g., when we would replace the pictures in Figure 17 with the words *apple*, *banana*, *lemon*, and *strawberry*). In Figure 8 we have a set diagram combined with a network, and set boundaries can also be overlaid on plots (e.g., when curves are drawn around clauses in a plot that use a particular tense, as in van der Klis et al. (2022)) and tables (Figure 5, see below).

Tables Figures 3, 4, 5, 11, 13, and 16 are tables. They all partition a piece of space into discrete cells. Figures 3 and 16 are straightforward tables, but the other tables require more explanation. The color space might be continuous, with its axes of hue, brightness, and saturation, but the representation that we have of it in Figure 5 is not a plot: the coordinates of brightness and hue have become the discrete rows and columns of a table. The box diagram in Figure 4 and the treemap in Figure 11 are structurally trees (with *cook* and *dog* as the roots, respectively), but diagrammatically tables (that is, partitions of a piece of space into cells). Figures 13 might be least recognizable as a table (even though it is called a matrix). What obscures the tabular nature of the Attribute-Value Matrix in 13 is the use of square brackets and the identity sign instead of lines around the cells.

As we already saw, hierarchical structures can be visualized by different types of diagrams (see also Schulz, 2011; Lima, 2014). They can be networks (Figure 7), tables (Figures 4 and 11), and set diagrams (examples in ethnobiological studies, Berlin (1992)). The notion of *tree* is not part of the primary graphic dimension of word meaning diagrams, although still important for characterizing the structure of certain diagrams, of course.

There are various secondary graphic parameters, relevant for word meaning diagrams, pertaining to orientation and layout, as we saw. Diagrams can also consist of multiple *panels*, (comparing different languages, different stages of a language, different words, ...). They can be web-based visualizations that allow the user to drill down from the general overview to the lexical details, opening up new panels and layers. Still, each part, panel or layer of such a dynamic, interactive visualization still belongs to one of the four graphic types.

4. The semantic dimension

For a typology of word meaning diagrams the graphic dimension does not suffice. We have no account yet for how diagrams of the same type (like networks) are being used in very different ways. Conversely, we lack a good understanding of how diagrams of different types (like networks and plots) can be used for roughly the same purpose (like semantic maps). We need a semantic dimension, not as an account of what diagrams in general mean, but how they show us something about word meanings. Two classical distinctions that are fundamental in the study of word meaning, are also important for word meaning diagrams: the distinction between *onomasiological* and *semasiological* and between *extensional* and *intensional*.

Onomasiology goes from meaning to words; it starts with a domain of meanings (say, different vessels) and it asks what those are called (*cup*, *mug*, *glass*, ...). Semasiology takes the reverse direction by asking what a particular word (like *cup*) means. When we apply this distinction to diagrams, we need to ask ourselves what the diagram is showing us something about: is it showing us the words we use for the meanings of a domain or is it showing us the meaning(s) of a particular word? Sometimes the relation between the diagram and the caption underneath can help. In an ideal onomasiological case the caption gives us a semantic domain (say, ‘names for vessels’) and the diagram gives us the words for that domain (say, *cup*, *mug*, *glass*, ...); if the caption gives us a word (e.g., *cup*) and the diagram gives us the meaning(s) for that word in one way or another, then the diagram is semasiological.

The extensional and intensional approach are about the way we characterize word meanings: in terms of instances or referents (e.g., pictures of individual vessels, different senses of the word *cup*, ‘vessel’, ‘prize’, ‘half pint’, ‘part of a bra’, ...) or in terms of how we define them (e.g. shape, material, function). Applied to diagrams the question is: what kind of semantic information is represented in the diagram about one or more words. If there are *instances* (senses, uses, referents, exemplars) in the diagram, as graphic ‘individuals’, then it is extensional, if no instances are shown, but there is information about *primitives* (attributes, conditions, features, relations), then the diagram is intensional. Note that an extensional diagram typically carries intensional information too, through the way it structures the instances.

Combining these two distinctions gives us four semantic types of diagrams (Table 1).

Table 1: Four semantic types of diagrams

	onomasiological	semasiological
extensional	<i>distributional diagram</i> shows how words of a domain distribute over its instances	<i>denotational diagram</i> shows how the instances of one word are structured
intensional	<i>relational diagram</i> shows how words of a domain are semantically related	<i>definitional diagram</i> shows how the definition of one word is structured

Table 2 shows the graphic type of network in all four semantic types.

Table 2: Four semantic types of networks

	onomasiological	semasiological
extensional	<i>distributional network</i> Figure 8 (e.g., classical semantic maps)	<i>denotational network</i> Figure 7 (e.g., radial networks)
intensional	<i>relational network</i> Figure 1 (e.g., semantic networks)	<i>definitional network</i> Figure 12 (e.g., frames)

In the top row, in the two types of extensional networks the nodes are instances, distinct senses or uses that words can have. The networks show how these are semantically related to each other (by similarity or specific semantic relations). The difference is whether the network of instances is the domain for multiple words (giving a distributional network, like Figure 8) or whether it belongs to one word (a denotational network, like Figure 7). In the bottom row, the two types of intensional networks have no instances and the nodes function in a different way. In a relational network the nodes correspond to words, semantically connected (e.g., by the relation of contrariety). Figure 1 is a relational network, and so is the tree of Porphyry, and many other classical and modern semantic networks of words. In a definitional network, both nodes and links represent analytical primitives, as in Figure 12, and more generally when the individual meaning of a particular word is visualized as a network, typically in frame semantics, but also elsewhere (e.g., in Schalley (2004)). In all four types it is the network that shows overall semantic structure, which is of course what a word meaning diagram is about.

The example of the network shows how one graphic type is used in semantically different ways, depending on how a semantic ‘ontology’ (with the three types of words, instances, and primitives) is related to the graphic structure of nodes and links. The reverse is also possible: a particular semantic type graphically expressed in different ways. In addition to distributional networks, we can also have distributional plots, distributional tables, and distributional set diagrams. These are all semantic maps, in different graphic formats. The distributional plot of Figure 9 organizes instances along axes and it shows the distribution of the words through graphic variables of shape and color. In the distributional table of Figure 5, the extensions of words are indicated by boundaries. Distributional tables are also common in the study of syncretisms in morphological paradigms, in the spirit of McCreight & Chvany (1991), where syncretisms are assumed to always involve contiguous cells: usually rectangular, but sometimes also L-shaped. More generally, it is in distributional networks, plots, and tables that constraints on the shape of word meanings play a role, like connectivity, contiguity, or convexity, definable in terms of an underlying structure that is visualized by the links, axes, and grid. This is different with the distributional subset diagram of Figure 10, which does not organize the instances in a way that is independent of the word boundaries.

All 4×4 types of diagrams exist (Table 3), although some might be more frequent, more natural and better-known than others.

Table 3: Word diagrams in the 4×4 typology

	network	plot	set diagram	table
definitional	12	6	2	13
denotational	7, 18	15	17	16
distributional	8	9	10	5
relational	1	14	19	3, 4, 11

For showing the intension of a single word, through a definitional diagram, not only networks are used, but also plots (Reichenbach’s *Simple Past* diagram in Figure 6), tables (Pustejovsky’s AVM of *bake* in Figure 13), and set diagrams (Euler’s representation of quantificational determiners, Figure 2).

In the examples of denotational diagrams, a table (Figure 16) and a set diagram (Figure 17) visualize in different ways how the extension of *fruit* is structured. The same information can also be expressed through a radial network that connects instances that differ

in one feature (Figure 18). A feature analysis of a large set of instances of a category can also be visualized through a plot, combining pile sorting (to let subjects determine the features) and multidimensional scaling (to reduce the many features to a small number of axes), e.g., for *fish* in Boster & Johnson (1989, p. 873).

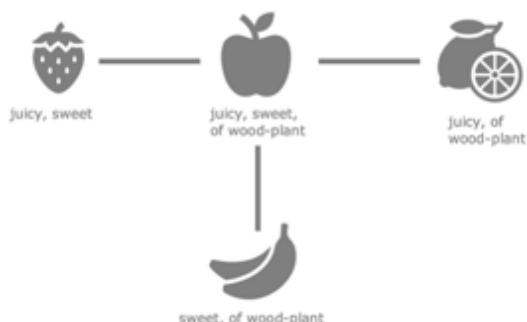


Figure 18: A denotational network for *fruit*

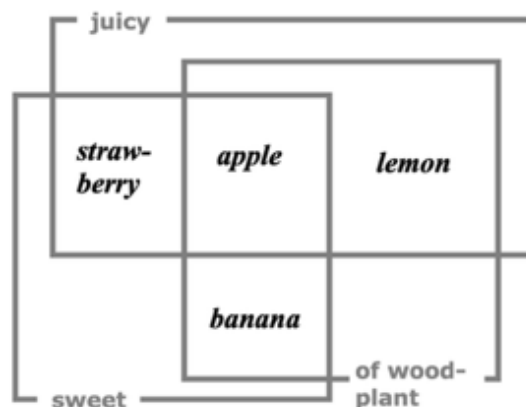


Figure 19: A relational set diagram for fruit words

In relational diagrams, the words themselves are the central players, as nodes in a network (Figure 1), points in a plot (Figure 14), or entries in a table (Figures 3, 4, 11). There is a surprising richness of relational tables, already in structuralist times: in addition to 3 and 4, the well-known diagrams associated to Trier and Hjelmslev are tables, that is, partitions of a semantic space into discrete cells. But tables also play a role in more recent visualizations of large WordNet hierarchies, not only as treemaps (Figure 11), but also as radial sunburst diagrams (e.g., Johannsen & Sandford Pedersen, 2011). No example was found of a relational set diagram, even though the idea is simple: showing how semantic features distribute over a set of words by encircling words that have the same semantic feature. The denotational diagram Figure 17 of the word *fruit* (semasiological + extensional) can be changed into a relational diagram for the domain of fruit (onomasiological + intensional) by replacing the pictorial instances with the corresponding words (Figure 19). The relations between the four fruit words are shown through their relative positions in a set diagram based on three semantic features.

One complication in the typology, is that relational diagrams (defined as onomasiological) can have a ‘focus’ word that allows them to function semasiologically. Take Figure 14, primarily classified as relational (onomasiological + intensional) because it plots the proximities between multiple words. However, because the word *gay* is foregrounded,

with the other words in the background, it shows us something about the meaning of *gay*, through its neighboring words.

There is a close relation between distributional diagrams and denotational diagrams. While distributional diagrams show the instances of multiple words, a denotational diagram shows the structure of the extension of one word. The denotational network of one word in one language can become the distributional network of multiple words when we expand, by including more words and more language. Conversely, in a distributional plot with many instances and words we could zoom in on the part carved out by one word and get a denotational plot.

Concluding this section, the introduction of a semantic dimension with two parameters, added to the four graphic types of Section 3, makes systematic sense of much of the variety that we find in word meaning diagrams. Of course, the variety of word meaning diagrams found in the literature is doubtlessly much richer than a 4×4 typology like this can capture, but this is a good starting point, with potential for wider applications, as Section 5 will explore.

5. Beyond the core typology

There is room for extensions in the graphic dimension (Section 5.1) and the semantic dimension (Section 5.2).

5.1 More on the graphic dimension

Pictures as a type of diagram were set aside at the beginning of this paper, to allow for a clear view on the core typology, but there is no denying they do play a role as word meaning diagrams, at least since the two sided *arbre* sign in Saussure (1916), in relation to Eskimo snow and Hopi time in Whorf (1940), in certain domains (body parts, spatial language), and prominently in cognitive linguistics, to represent image schemas, in the work of Johnson, Lakoff, Langacker, Talmy. How could they fit into the typology, if we allow pictures as a fifth type of graphic? Do we then find definitional, denotational, distributive, and relational pictures?

A distinction between semasiological and onomasiological functions is useful. Figure 20 is semasiological, because we go from the one word *dent* to a picture intended to represent its meaning. Figure 21, on the other hand, is onomasiological, because we go from a domain, the human body, shown by its picture, to words for its parts.

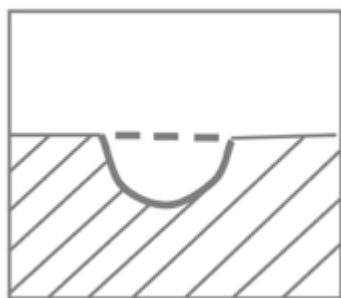


Figure 20: *Dent* (Langacker, 1986, p. 23)

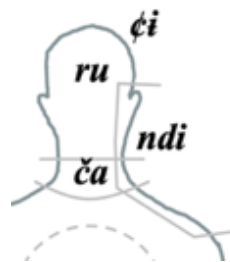


Figure 21: Tarascan body parts (partial, Friedrich & Krauss, 1969)

For applying the extensional/intensional distinction to pictures, the status of the graphic individuals in pictures is relevant. The graphic individuals of networks, plots, set diagrams, and tables can stand for instances (the extensional case) or for primitives (the intensional case). Parts of pictures can also be significant graphic individuals, like the boundary and inside of the dent in Figure 20 and the body part regions in Figure 21. The different parts in Figure 20 are clearly not different instances of the word *dent*: they are the things that define a dent, like semantic components, only pictorially represented, making this an intensional diagram, and hence a definitional picture. The word *ča* in Figure 21 corresponds to the throat of the body and *ndi* to the shoulder. These two picture parts are not two different instances of the same word; they are the regions that uniquely characterize the meaning of these words. This diagram too is intensional, more specifically a relational picture. Like other relational diagrams, Figure 21 shows how a set of words (body part terms, in this case) hangs together. Other diagrams use axes, boundaries, a grid, a network for that, but a relational picture uses the spatial structure of the picture.

Are there any *extensional* pictures, pictures in which different parts represent alternative instances of a word? The domain where we find examples are spatial expressions, like prepositions. In a typical extensional approach, Logan & Sadler (1996) instructed subjects to “Draw an X «*P*» the box” on a sheet of paper for various prepositions «*P*» and visualized the results. Figure 22 shows the centers of the X’s for «*P*» = *away from* (with round dots). Figure 23 adds points (little squares) for *near*.¹⁴

¹⁴ Logan & Sadler have 12 separate panels for 12 prepositions, including *away from* and *near*. They do not have something corresponding to Figure 23, which is constructed for the purposes of this paper.



Figure 22: Points for *away from* (Logan & Sadler, 1996, p. 508)

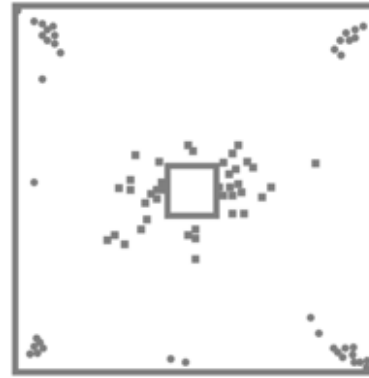


Figure 23: Points for *near* added

These diagrams are pictures, not plots, because the axes are not abstract dimensions, but physical ones. The diagrams are extensional, because the dots represent alternative instances of prepositions. What we get then in Figure 22 is a denotational picture, showing the spatial structure of the extension of *away from*. Figure 23 is a distributional picture, comparing the spatial distributions of *away from* and *near*. Notice the similarity of the distributional picture to the distributional plot in Figure 9 (which shows the distributions of French prepositions in an MDS-derived space based on parallel corpora). The underlying spaces are very different, of course, but they are both onomasiological and extensional, and hence distributional.

Word meaning pictures can be denotational, distributional, and relational, but they are most frequently of the definitional type, especially in cognitive linguistics, where they represent image schemas. These diagrams tend to be schematic and no longer purely iconic. In the force dynamic diagrams of Talmy (1985), for instance, agent and patient are symbolized by (arbitrarily) different shapes and their interaction by additional symbols (like + and >). The *Diagrammatic Image Schema Language* of Hedblom et al. (2025) is an attempt to systematize such force dynamic diagrams. These diagrams push the boundaries of what counts as pictorial in a way that deserves further study.

The definition of graphic representations that we started this paper with also encompasses texts. Could they constitute an additional type for word meaning diagrams? The question is not entirely theoretical. Fitzmaurice et al. (2017, pp. 34–35) show historical texts with strong collocates of the word *valour* (itself underlined) color coded. In that way we get an impression of the semantic field of a word projected on an actual text. We get an idea of the meaning of *gay* in the 1900s through the neighboring words in a plot (Figure 14), and we get an idea of the meaning of *valour* in the 1500s through certain words in the text around it.

However, in contrast to graphic structure of the plot, the graphic structure of the text does not really play a role in ‘seeing’ the structure of the semantic field of *valour*. Distances and shapes do not play a role; we need to read the content of the text. Even though the annotated texts of Fitzmaurice et al. are diagrammatic, it is unclear whether they are word meaning diagrams of some sort.

A final additional type to consider is the *cloud*, which is here taken as a set of graphic components (typically words) without any graphic organization, apart from variation in size and color).¹⁵ The *word cloud* shows a set of words arbitrarily positioned on the page. No links, no boundaries, no axes, no grid. Although primarily used for text analysis, word clouds also appear in recent literature about word meanings, for instance in Koptjevskaja-Tamm & Sahlgren (2014, p. 246) to show the collocates of *warm* found through word-space modelling in a corpus. Even though a word cloud by definition does not show structure, we could see this as a ‘relational cloud’, a diagram that shows something about the meaning of the word *warm* by just showing other words. Other types of clouds (in particular with instances) are theoretically possible (like a ‘denotational cloud’ showing the instances of a word) but not encountered yet.

The graphic dimension is not just about types, it is also about how different types can be combined with each other. We already saw how word meaning diagrams can have layers of different types, but there are also other ways of combining:

- **Network & plot** Croft (2012) represents verb meanings using two-dimensional plots (time × quality) for subject and object and a causal arrow between those plots, making these diagrams networks that embed plots.
- **Network & set diagram** In the semantic modality map of Van Der Auwera & Plungian (1998) modal meanings are sometimes related through links and sometimes through nested circles, a hybrid of network and set diagram.
- **Picture & plot** Langacker (1986, p. 25) has a sequence of pictures along a temporal axis to represent *across*, with a circle for the ‘trajector’ at various positions with respect to a square for the ‘landmark’.
- **Picture & plot** In Logan & Sadler’s experimental study of prepositions, there are also diagrams (1996, pp. 510–511) that show the acceptability rate of each position around

¹⁵ This notion of cloud is different from that of Montes (2021), which refers to tokens of a word in a denotational semantics plot.

a box for a particular preposition three-dimensionally, through relief, by plotting the acceptability rates as heights, orthogonal to the two-dimensional plane of the picture.

Thus, by allowing combinations of basic graphic types, the typology helps to understand the open-ended, multidimensional design space that scholars working on word meaning are using and expanding.

5.2 More on the semantic dimension

The two parameters of the semantic dimension, onomasiological/semasiological and extensional/intensional, define a useful core for understanding the variety of word meaning diagrams, but there are various aspects of word meaning diagrams that show that we need to go beyond these two parameters and beyond how they are defined or applied.

Word-less diagrams There can be word meaning diagrams that do not show any words. The semantic analysis of a set of words can lead to a ‘bare’ network or plot of meanings. This is almost always an option with semantic maps and with most distributional diagrams: what we get without words is a perspective on the structure of a semantic domain or conceptual space, typically as a network or plot. One example of this type from before the advent of semantic map is the network of sensory domains in Williams’ (1976) study of synesthetic adjectives (e.g., *sharp* sound, taste, ...). More recent examples are the networks of concepts based on the CLICS database (Mayer et al., 2014) and the circular network showing metaphorical mappings between semantic domains found in the Oxford English Dictionary (W. Anderson et al., 2016), although in these more recent interactive diagrams we can always call up the words with a click of the mouse. We need to take *onomasiology* broader, not just for the sake of diagrams, but for understanding much work in cognitive science more generally: starting with a domain of meanings of some sort, the goal is not just to find out how those meanings are named, but also what those naming patterns (like the colexifications) reveal about how the meanings are organized, maybe even independently of language.

Formal Concept Analysis In networks, the core typology has nodes exclusively of one of the three types (words, instances, primitives). Word meaning diagrams within Formal Concept Analysis (FCA) show that this is too strict. FCA provides a generic way of visualizing a set of objects with attributes as a lattice, a Hasse network (Ganter & Wille, 1999). This method has been applied to multilingual dictionaries (Janssen, 2002), lexical databases (Priss, 2005), and semantic maps (Ryzhova & Obiedkov, 2017). Figure 24 has six words for horses as objects and Figure 25 has instances of *triangle* as objects. Ryzhova &

Obiedkov (2017) have an FCA diagram for ‘sharp’ in which the objects are words like English *sharp* and the attributes are instances (sharp form, sharp point, sharp edge).¹⁶

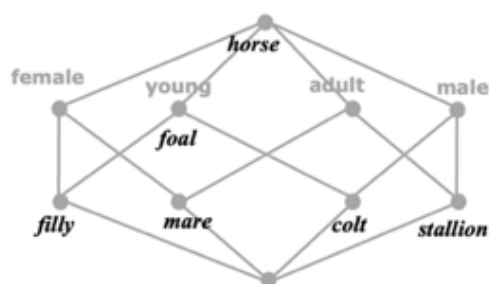


Figure 24: FCA lattice for words for horses
(Janssen, 2002, p. 38)

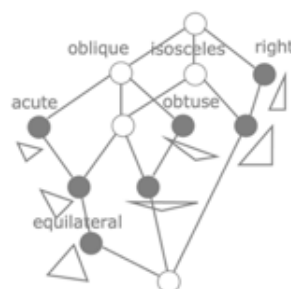


Figure 25: FCA lattice for *triangle* (Priss, 2021, p. 288)

In these diagrams the objects (words in Figure 24 and in Ryzhova & Obiedkov, instances in Figure 25) are ‘dominated’ by the attributes that they have (primitives in Figures 24 and 25, instances in Ryzhova & Obiedkov). What makes FCA diagrams special, in comparison to other word meaning networks, is that they have two very different types of elements as nodes in the same network while the lines always represent the same partial order. Still, we would like to say that Figure 24 is a relational network (onomasiological + intensional), that Figure 25 is a denotational network (semasiological + extensional), and that the diagram of Ryzhova & Obiedkov is a distributional network (onomasiological + extensional). There is enough flexibility in the relation between graphic individuals and the semantic ‘ontologies’ for the typology to also include FCA diagrams.¹⁷

Exemplary diagrams Denotational diagrams show how a set of instances of one word is structured. Could there also be diagrams that show one instance, as an example? This is of course what often happens with pictures. Instead of giving a very schematic, general picture, as a definitional diagram, such an *exemplary picture*, as we might call it, can be more mimetic. This is what we see in some dictionaries, with pictures exemplifying words. An additional type of exemplary diagrams also creates room for the kind of networks found in the semantics of *each other* (at least since Fiengo & Lasnik, 1973, p. 449), to represent particular

¹⁶ A node stands for a *formal concept*, a pair (E, I) where E is the set of objects that have all the attributes in I and I is the set of attributes that all the objects in E have. $(E, I) \leq (E', I')$ when $E' \subseteq E$ and $I \subseteq I'$. The version of Figure 25 uses open circles for formal concepts that have an empty set.

¹⁷ There are no *definitional* FCA diagrams. FCA is always about how a set of objects is classified by a set of attributes (words by features, instances by features, instances by words or vice versa). In word meaning diagrams, however, there are only elements of one type: the primitives that serve as building blocks of definitions.

situations that make a sentence like *the men are hitting each other* true: the men are nodes and arrows shows who is hitting who. This is neither a definitional network (because it does not cover all possible situations) nor a denotational network (because it does not show the variety of possible situations), but an exemplary network.

Diachronic diagrams The most obvious parameter to add to the typology is diachrony. A diachronic word meaning diagram, like Figure 14, can show how word meanings change. The study of word meaning change is rapidly expanding, thanks to the advent of new computational methods, bringing along attention for different visualization methods in that domain (e.g., Rohrdantz et al., 2011; Zhang et al., 2015; Theron & Fontanillo, 2015; Xu & Crestani, 2017; Jatowt et al., 2021; Kazi et al., 2022). The variety of diachronic word meaning diagrams is enormous and worth a separate study, building on the typology presented here. One question is then how the dimension of time combines with the types of this typology. What comes to mind are side-by-side diagrams for two different times, directed links in semantic map networks, arrows in word spaces, plotting semantic values along a horizontal time axis.

6. Conclusion

The core of this paper is a two-dimensional typology of word meaning diagrams. The graphic dimension goes beyond a mere list of types to a systematic and comprehensive constellation of graphic types, taking into account possible building blocks and combinatorial mechanisms. The semantic dimension puts familiar distinctions to a new use, creating four basic semantic types that cross-cut the graphic types. There is a many-to-many relation between graphics and semantics in the ‘visual language’ of word meaning diagrams: one graphic type can have very different uses and very different graphic types can still be of the same semantic type.

A typology like this, giving insight in the design space of word meaning diagrams, is not only practically useful in showing ‘diagrammers’ what is possible, but it also allows for foundational and historical insight in the study of word meaning in all its historical, theoretical, and methodological variety. How do older diagrams compare to newer ones and what does this show about the development of lexical semantics? How does the use of diagrams vary across different approaches (e.g., cognitive versus formal semantics)? How does the structure of diagrams connect to the structure of lexical semantic theories? How can computational methods, not only for ‘drawing’ new diagrams on the basis of text, but also for ‘reading’ diagrams, play a role? There are enough loose ends and opportunities remaining, and hopefully the typology offered in this paper can help addressing those.

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